CS4433: INDIVIDUAL ASSIGNMENT

Student Number: **202321241**

4. Requirements Engineering

4.1 Stakeholder Analysis and Conflict Resolution

4.1.1 Stakeholders and Their Interests

The system has five stakeholders, each with a distinct interest in how the system behaves.

The Principal requires a system that is simple enough for non-technical staff to operate, and requires that marks remain hidden from students until they have been checked and approved by the Head of Department.

The Lecturer requires marks to be released to students as soon as they have been entered, and requires the discretion to grant individual students an extension when there is a good reason for one.

The Head of Department requires an automated late-penalty policy (10% of the available mark deducted for each day late, up to three days, after which the mark is zero) and requires full visibility of every mark in the department before release.

The Student Representative requires students to be able to correct and resubmit their work before the deadline, requires written feedback rather than a bare numeric mark, and requires that late work not simply be rejected outright.

The IT Officer requires the system to operate fully offline, requires stored passwords to be protected, and requires the system to remain responsive with the full complement of approximately 800 students using it.

The Administrator is a system user rather than an interviewed stakeholder. Its requirements are derived from the case-study description (user, department, course and enrolment management).

4.1.2 Conflicts and Ambiguities

Three genuine conflicts and two ambiguities were identified among these stakeholder statements. Each is resolved below through a specific design decision, so that the resolution can be traced directly into the Software Requirements Specification and later into the class design.

Conflict 1: Timing of mark release**.** The Principal requires marks to be withheld from students until the Head of Department has approved them, while the Lecturer requires marks to be visible to students as soon as they are entered. These requirements are mutually exclusive as stated, and no reinterpretation removes the conflict. The Principal's requirement is given priority, because departmental moderation exists precisely to catch marking errors and inconsistencies before a student sees a result, and reversing an already-visible mark carries a reputational and fairness cost that outweighs the Lecturer's convenience. The system therefore enforces the submission life cycle of Submitted or Late, Marked, Moderated, and Released, and a mark is only visible to the student once the submission reaches Released. The cost to the Lecturer is mitigated rather than eliminated:

marks can be entered the instant marking is complete with no technical delay, moderation requests are queued for the Head of Department immediately on entry, and the Lecturer can see the moderation status of every mark they have entered, so unmoderated marks are visible for tracking purposes even though they are not yet visible to the student.

Conflict 2: Extensions versus a strict penalty policy. The Lecturer requires discretion to grant individual extensions, while the Head of Department requires the late-penalty policy to be strict and automatic. These are reconciled by modelling an extension as a record that changes the effective deadline for a specific student and assignment, rather than as a mechanism that allows the Lecturer to alter a computed penalty. The penalty calculation always runs mechanically against whichever deadline is currently in effect, so the Head of Department's requirement that the policy be automatic and non-negotiable is preserved. The Lecturer's discretion is scoped narrowly to adjusting the deadline itself, not to overriding the mark that the policy produces. This scoping is a deliberate design decision and is presented in the report as such.

Conflict 3: Rejection versus leniency for late work. The Head of Department's statement, that work more than three days late receives zero, appears at first to reject late work, which seems to conflict with the Student Representative's requirement that late work not simply be rejected. These statements are compatible once submission acceptance is distinguished from mark calculation. The system always accepts a late submission and allows it to be marked and given written feedback, satisfying the Student Representative's requirement. The penalty policy then determines the numeric outcome of that submission, which may be zero once the grace period has passed, satisfying the Head of Department's requirement. Acceptance of a submission and the score it eventually receives are therefore treated as two independent concerns.

Ambiguity 1: Numeric mark versus written feedback. No stakeholder statement other than the Student Representative's mentions feedback text, so this is treated as an underspecified but binding requirement rather than a true conflict. Every released mark must carry an associated feedback text field in addition to its numeric score, and the system prevents a mark from being released while that field is empty.

Ambiguity 2: Scope of resubmission. The Student Representative's statement that students should be able to resubmit corrected work before the deadline does not specify how many times resubmission is allowed or how it interacts with the late-penalty clock. This is resolved by an explicit assumption: resubmission is permitted only while the current submission has not yet been marked and only before the effective deadline for that student and assignment. Each resubmission replaces the previously submitted file, but the audit trail retains a record of the submission history. This assumption is stated explicitly in the Software Requirements Specification, since the project brief does not specify it.

4.2 Software Requirements Specification

4.2.1 Purpose, Scope and Structure

This section specifies the requirements of the Moderated Learning Management System, release v1.0. It follows the structure recommended by ISO/IEC/IEEE 29148: purpose and scope, definitions, assumptions and business rules, functional requirements, non-functional requirements, and

constraints. The system lets an Administrator set up users, departments, courses and enrolments; lets Lecturers create assignments, grant extensions and mark submissions; lets Students submit, resubmit and view released marks; and lets Heads of Department moderate and release marks, with every mark change recorded in an audit log.

Each requirement is written as a "shall" statement and has a unique ID, a MoSCoW priority (M Must, S Should, C Could), a source, the use cases it relates to, and a verification method (T test, I inspection, D demonstration, A analysis). The source column uses these codes: PR Principal, LE Lecturer, HD Head of Department, SR Student Representative, IT IT Officer, CS case study, Brief project brief, and C1 to C3 and A1 and A2 for the conflicts and ambiguities resolved in Section 4.1.2. The Administrator's requirements are taken from the case study, because that role has no stakeholder statement.

4.2.2 Definitions

Term	Meaning
Assignment deadline	The deadline the Lecturer sets when creating an assignment.
Extension	A record that gives one student a later deadline for one assignment.
Effective deadline	The deadline that applies to one student for one assignment: the assignment deadline, or the student's extension deadline if one exists.
Raw mark	The mark entered by the Lecturer, or changed by the Head of Department, before any late penalty.
Final mark	The raw mark after the late penalty. This is the mark the student receives.
Days late	The number of whole days, rounded up, between the effective deadline and the submission time.
Submission status	One of Submitted, Late, Marked, Moderated or Released.
Moderation	The Head of Department's review of a Marked submission, ending in approval (with or without a change to the raw mark) or return to the Lecturer.
Release	The Head of Department's action that makes a Moderated mark and its feedback visible to the student.
Server time	The time from the server's clock in UTC. It is the only time source used by the system.

4.2.3 Assumptions and Business Rules

The project brief and the stakeholder statements leave some points open. The assumptions below are stated so that they can be reviewed, and the business rules are the rules the system enforces. The use case specifications and the tests refer to them by ID.

ID	Assumption
AS1	All timestamps come from the server clock in UTC. The client's clock is never used.
AS2	A submission is one file of at most 5 MB.
AS3	"10% of the available mark" means 10% of the assignment's maximum mark, not 10% of the raw mark.
AS4	Each course belongs to exactly one department. A department has one Head of Department. A course may have several Lecturers, and a student may be enrolled in several courses.
AS5	The school has about 800 students. Load tests use up to 50 simultaneous users, which is an assumption about peak use and not a figure given in the brief.
AS6	Correcting a mark after release is outside v1.0.

ID	Business rule
BR1	Effective deadline. The effective deadline for a student and assignment is the assignment deadline, or the student's extension deadline if one exists. An extension must be later than the current effective deadline, and a newer extension replaces an older one. An extension changes the deadline only. It never changes the penalty rule or its result.

ID	Business rule
BR2	Acceptance. A submission is on time if its submission time is not after the effective deadline. Otherwise it is Late. Late submissions are always accepted, marked and given feedback.
BR3	Late penalty. Let L be the submission time minus the effective deadline, in seconds. If L is 0 or less, days late d = 0. Otherwise d = ceil(L / 86 400). d = 0: final mark = raw mark. d = 1, 2 or 3: final mark = max(0, raw mark - d x 0.10 x maximum mark). d of 4 or more: final mark = 0. Boundaries: exactly 24 hours late gives d = 1; 24 hours and 1 second gives d = 2; exactly 72 hours gives d = 3; 72 hours and 1 second gives d = 4 and a final mark of 0. Example: maximum mark 50, raw mark 40, submitted 1 day 2 hours late. d = 2, penalty = 10, final mark = 30. The system calculates the penalty. No user can edit or override it. A Head of Department adjustment changes the raw mark only.
BR4	Resubmission. A student may resubmit only while the current submission has status Submitted, which means before the effective deadline and before marking. The new file replaces the current one and the earlier one is kept in the submission history. A Late submission cannot be replaced.
BR5	Life cycle. Status changes only as Submitted or Late, then Marked, then Moderated, then Released. A submission returned by the Head of Department stays Marked and is flagged Returned until it is re-marked. No other change is allowed, and marks cannot change once the submission is Moderated.
BR6	Visibility. A student sees a mark and its feedback only when the submission is Released. Before that the student sees only the submission status (Submitted, Late or Awaiting results).
BR7	Feedback. A submission cannot be marked without written feedback and cannot be released while its feedback is empty.
BR8	Audit. Every creation or change of a raw or final mark is recorded with who, when, old value, new value and reason, in the same transaction as the change. The reason is "Initial mark" for first marking and is mandatory for re-marking and for Head of Department adjustments.
BR9	Role scope. A Lecturer acts only on courses they are assigned to. A Head of Department acts only on courses in their own department. A Student sees only their own submissions and enrolled courses.
BR10	Extension after submission. If an extension is granted to a student whose submission is not yet marked, the status and the penalty are re-evaluated against the new effective deadline.

4.2.4 Functional Requirements

ID	Requirement	Pri.	Source	Use case	Ver.
FR01	The system shall authenticate each user by a unique username and password and reject invalid credentials without revealing which of the two was wrong.	M	CS, IT	UC01	T
FR02	The server shall authorise every request against the authenticated user's role and reject any action outside that role, for all four roles, whatever the client displays.	M	CS, PR	All	T
FR03	The system shall lock an account for 5 minutes after 5 consecutive failed logins and end a session after 15 minutes of inactivity.	S	IT	UC01	T
FR04	The Administrator shall create, edit and deactivate user accounts, each with a role of Student, Lecturer, Head of Department or Administrator. A deactivated user cannot log in, and accounts that hold marks or submissions are never deleted.	M	CS	UC02	T
FR05	The Administrator shall reset a user's password, and every user shall be able to change their own password.	S	IT	UC02, UC01	T
FR06	The Administrator shall create and edit departments and assign one Head of Department to each.	M	CS	UC03	T
FR07	The Administrator shall create and edit courses, each belonging to exactly one department.	M	CS	UC04	T
FR08	The Administrator shall assign one or more Lecturers to a course.	M	CS	UC05	T
FR09	The Administrator shall enrol students in courses and remove them from courses.	M	CS	UC06	T

ID	Requirement	Pri.	Source	Use case	Ver.
FR10	A Lecturer shall create an assignment for a course they are assigned to, giving a title, description, deadline and a maximum mark greater than zero.	M	CS, LE	UC07	T
FR11	A Lecturer shall grant a student an extension for an assignment by setting a new deadline later than that student's effective deadline, with a reason (BR1, BR10).	M	LE, HD, C2	UC08	T
FR12	A Lecturer shall view all submissions for their assignments, with student, status, submission time and effective deadline.	M	CS	UC09	T
FR13	A Lecturer shall mark a submission by entering a raw mark from 0 to the maximum mark and non-empty written feedback. On saving, the system shall set the status to Marked and add the submission to the Head of Department's moderation queue.	M	CS, LE, C1, A1	UC10	T
FR14	A Student shall view their enrolled courses and, for each, its assignments with the effective deadline that applies to that student.	M	CS	UC12	T
FR15	A Student shall submit one file for an assignment of an enrolled course. The system shall accept it before or after the effective deadline, record the server time and set the status to Submitted or Late (BR2).	M	CS, SR, C3	UC13	T
FR16	A Student shall resubmit while the current submission has status Submitted (BR4). The system shall keep the earlier file in the submission history.	M	SR, A2	UC14	T
FR17	A Student shall see their mark and feedback only when the submission is Released. Before that the system shall show only the submission status (BR6).	M	PR, C1	UC15	T
FR18	The system shall calculate the days late, penalty and final mark automatically using BR3. No user shall be able to change the penalty.	M	HD, C2, C3	UC21	T
FR19	The system shall enforce the submission life cycle in BR5 and reject any other status change.	M	CS	All	T
FR20	A Head of Department shall view the raw mark, final mark, status, feedback and Lecturer of every submission in the courses of their department, and of no other department.	M	HD	UC16	T
FR21	A Head of Department shall moderate a Marked submission by approving it, which sets the status to Moderated, or by changing the raw mark with a mandatory reason and then approving it.	M	PR, HD	UC17, UC18	T
FR22	A Head of Department shall release Moderated submissions, one at a time or all of an assignment at once, which sets the status to Released. The system shall refuse to release a submission whose feedback is empty.	M	PR, A1	UC17	T
FR23	The system shall record every creation or change of a mark in an audit log holding who, when, old value, new value and reason (BR8).	M	CS	UC22	T
FR24	A Lecturer shall view the moderation status (Marked, Returned, Moderated or Released) of every mark they entered.	S	LE, C1	UC11	D
FR25	A Head of Department shall return a Marked submission to its Lecturer with a mandatory reason. The submission stays Marked and flagged Returned until it is re-marked.	S	HD	UC19	T
FR26	A Head of Department shall view the audit log entries for marks in their department.	S	HD, CS	UC16	D
FR27	A Student shall see, with a released mark, the raw mark, the days late, the penalty and the final mark.	S	SR, HD	UC15	T
FR28	The system shall show a Student, at login, a notice of marks released since their last login.	C	SR	UC15	D

4.2.5 Non-Functional Requirements

Every non-functional requirement states a measurable limit and the way it will be checked. The load figures follow assumption AS5.

ID	Category	Requirement	Pri.	Source	Ver.
NFR01	Performance	Login shall complete within 2 seconds at the 95th percentile with 800 student accounts and 50 staff accounts stored and 50 concurrent users, on a server with 2 CPU cores and 4 GB of RAM.	M	IT	T
NFR02	Performance	Listing a student's assignments, a Lecturer's submissions or a department's marks shall complete within 2 seconds at the 95th percentile under the load in NFR01.	M	IT	T
NFR03	Capacity	The system shall accept 800 submissions within a 10-minute window with no submission lost or corrupted, checked by comparing a checksum of every stored file.	M	IT	T
NFR04	Security	Passwords shall be stored only as salted hashes, using Argon2id from libsodium with a unique salt for each user. Inspection of the database and logs shall find no plain-text password.	M	IT	I, T
NFR05	Security	Every SQL statement shall use bound parameters, and no SQL shall be built by string concatenation. Twenty SQL injection test strings shall be rejected with no change to the data.	M	CS, IT	I, T
NFR06	Availability	All functions shall work with no connection beyond the school network. All system tests shall pass with the server's external network connection disabled.	M	IT	T
NFR07	Robustness	Invalid input or a database error shall not stop the server or the client. A suite of at least 30 invalid-input and fault cases shall produce zero crashes, and each case shall return an error message.	M	CS	T
NFR08	Integrity	Each operation that changes data shall be a single transaction. After the server process is killed during a write, the database shall pass SQLite's integrity check and hold no partial record.	M	IT	T
NFR09	Auditability	The audit record and the mark change shall be written in the same transaction. In testing, every mark change shall have exactly one audit record, and no function shall exist to edit or delete audit records.	M	CS	T, I
NFR10	Usability	A first-time Lecturer using only the user manual shall create an assignment within 3 minutes, and every function of each role shall be reachable within three menu selections from that role's main screen. Both are checked with 3 non-technical users.	M	PR	D
NFR11	Resource use	The client shall run on a PC with 2 GB of RAM and a dual-core CPU using no more than 100 MB of memory. The server shall use no more than 300 MB with 800 students' data.	M	IT	T
NFR12	Maintainability	Automated tests shall cover at least 70% of the lines of the business-logic layer, and the actual figure shall be reported.	S	Brief	A
NFR13	Recoverability	An Administrator shall be able to back up and restore the database with one command each. A backup of 800 students' data shall take less than 1 minute, and a restored copy shall pass the integrity check.	S	IT	T

4.2.6 Constraints and Exclusions

The system is written in C++17 and stores data in SQLite. It runs on a single school computer or the school network with a console or simple desktop client, and it does not depend on the internet. Online payment, video conferencing, automated grading, plagiarism detection and web or mobile interfaces are out of scope, as the project brief states. The following are not in v1.0: correcting a mark after release (AS6) and notification by email. Items marked C are built only after all Must and Should requirements are complete and tested.

4.3 Use Case Model

4.3.1 Actors

The system has four actors that match the four user roles in the case study. They share one abstract actor, User, which stands for any person who logs in. The four roles generalise User, so Log in is associated with User once instead of with each role.

Actor	Goals
User (abstract)	Log in. All four roles inherit this.
Student	View enrolled courses and assignments; submit and resubmit work; view released marks and feedback.
Lecturer	Create assignments; grant individual extensions; mark submissions and write feedback; follow the moderation status of the marks they entered.
Head of Department	Review the marks of every course in the department; moderate marks; release approved marks to students.
Administrator	Manage user accounts, departments and courses; assign Lecturers to courses; enrol students.

4.3.2 Use Case Diagram

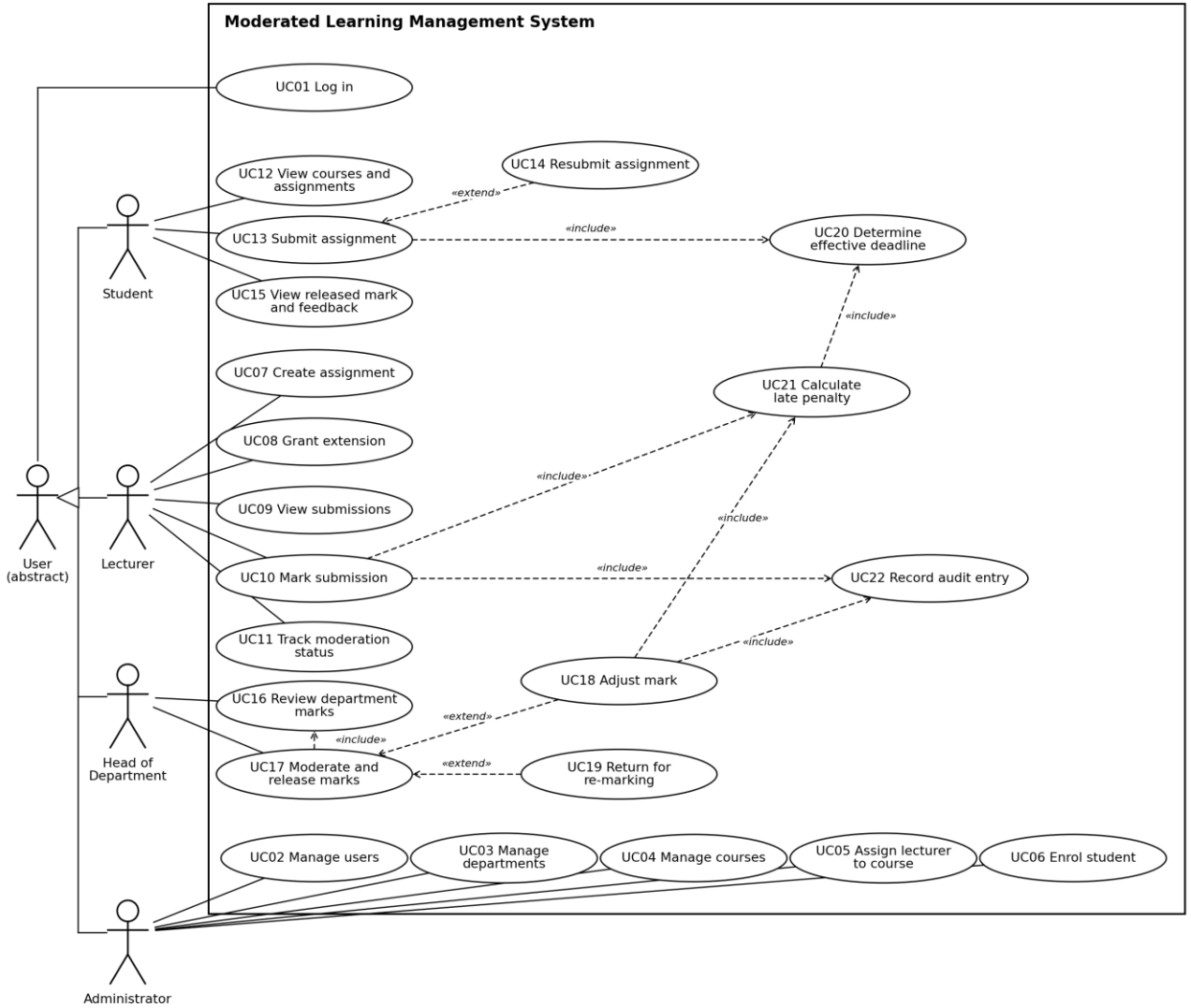


Figure 4.1: Use case diagram. UC20 to UC22 are shared use cases that the system performs for other use cases.

4.3.3 Use Case Catalogue and Relationships

Include is used where a use case always needs another behaviour: Submit assignment always needs the effective deadline (UC20), Mark submission always calculates the penalty (UC21) and records an audit entry (UC22), and Moderate and release marks always starts from the department's marks (UC16). Extend is used where the behaviour happens only under a condition: Resubmit assignment extends Submit assignment when the current submission is still Submitted, and Adjust mark and Return for re-marking extend Moderate and release marks when the Head of Department decides to change or return a mark. Generalisation is used for the actors.

ID	Use case	Primary actor	Relationships	Requirements
UC01	Log in	User (all roles)		FR01, FR03, FR05
UC02	Manage users	Administrator		FR04, FR05
UC03	Manage departments	Administrator		FR06
UC04	Manage courses	Administrator		FR07
UC05	Assign lecturer to course	Administrator		FR08
UC06	Enrol student	Administrator		FR09
UC07	Create assignment	Lecturer		FR10
UC08	Grant extension	Lecturer		FR11
UC09	View submissions	Lecturer		FR12
UC10	Mark submission	Lecturer	include UC21, UC22	FR13, FR18, FR23
UC11	Track moderation status	Lecturer		FR24
UC12	View courses and assignments	Student		FR14
UC13	Submit assignment	Student	include UC20	FR15
UC14	Resubmit assignment	Student	extend UC13	FR16
UC15	View released mark and feedback	Student		FR17, FR27, FR28
UC16	Review department marks	Head of Department		FR20, FR26
UC17	Moderate and release marks	Head of Department	include UC16	FR21, FR22
UC18	Adjust mark	Head of Department	extend UC17; include UC21, UC22	FR21, FR23
UC19	Return for re-marking	Head of Department	extend UC17	FR25
UC20	Determine effective deadline	System	included by UC13, UC21	FR11, FR14
UC21	Calculate late penalty	System	include UC20; included by UC10, UC18	FR18
UC22	Record audit entry	System	included by UC10, UC18	FR23

4.3.4 Fully Dressed Use Case Specifications

Three use cases are specified in full: Submit assignment, Mark submission, and Moderate and release marks. Together they cover the path from a student's work to a released mark, and between them they exercise the late, resubmission, penalty, audit and moderation rules and the main exception paths that the system tests will use.

Use case UC13: Submit assignment

Primary actor	Student
Stakeholders and interests	Student: the work is accepted, a receipt is given, and late work is treated fairly. Lecturer: receives the work. Head of Department: the late-penalty rule is applied the same way every time. Student Representative: late work is not simply rejected.
Trigger	The student chooses to submit work for an assignment in their assignment list.

Preconditions	P1 The student is logged in (UC01). P2 The student is enrolled in the course. P3 The assignment exists. P4 The student has no submission for this assignment, or the current submission has status Submitted (A1).
Main flow	1. The student selects an assignment and chooses Submit. 2. The system determines the effective deadline for this student and assignment (UC20) and shows the title, the effective deadline and the 5 MB limit. 3. The student selects a file and confirms. 4. The system checks that a file was chosen and that it is no larger than 5 MB. 5. The system records the current server time as the submission time. 6. The submission time is not after the effective deadline, so the system sets the status to Submitted. 7. The system stores the file, the submission time and the status in one transaction. 8. The system shows a receipt with the submission time and status.
Alternative flows	A1 Resubmission (UC14, at step 1). The student already has a submission with status Submitted. The system warns that the new file will replace it. On confirmation the flow continues from step 3, and at step 7 the new file becomes the current submission while the earlier one is kept in the submission history. A2 Late submission (at step 6). The submission time is after the effective deadline. The system sets the status to Late and accepts the submission. The receipt says that a late penalty will apply at marking (10% of the maximum mark for each day, up to three days, then zero). The flow continues at step 7. A3 Cancel (at steps 1 to 3). The student cancels and nothing is stored.
Exception flows	E1 Invalid file (step 4). No file was chosen or the file is over 5 MB. The system shows what is wrong and returns to step 3. E2 Not permitted (step 1). The student is not enrolled in the course. The server rejects the request and stores nothing. E3 Resubmission not allowed (step 1). The current submission is Late, Marked, Moderated or Released. The system tells the student it can no longer be replaced, and the use case ends. E4 Storage or database error (step 7). The transaction is rolled back and any earlier submission is unchanged. The system tells the student to try again. No partial record remains. E5 Session expired (any step). The system returns to login and stores nothing.
Postconditions	Success: exactly one current submission exists for the student and assignment, with the file, the submission time and the status Submitted or Late. Replaced files remain in the history. The Lecturer can see the submission. Failure: stored data is unchanged.
Business rules	BR1, BR2, BR4, AS1, AS2. BR3 is applied later, when the submission is marked.
Requirements	FR15, FR16, FR19, NFR03, NFR05, NFR07, NFR08

Use case UC10: Mark submission

Primary actor	Lecturer
Stakeholders and interests	Lecturer: marking is quick and the result is clear. Head of Department: the penalty is applied automatically and every mark can be traced. Student: receives written feedback. Principal: no student sees a mark before it is moderated.
Trigger	The Lecturer selects a submission to mark from the list of submissions (UC09).
Preconditions	P1 The Lecturer is logged in and is assigned to the course. P2 The submission has status Submitted or Late, or is Marked and flagged Returned.
Main flow	1. The Lecturer opens a submission from the list. 2. The system shows the file, the submission time, the status, the effective deadline and the maximum mark. 3. The Lecturer enters a raw mark and written feedback.

	4. The system checks that the raw mark is from 0 to the maximum mark and that the feedback is not empty. 5. The system calculates the days late, the penalty and the final mark (UC21) and shows the raw mark, days late, penalty and final mark. 6. The Lecturer confirms. 7. In one transaction the system stores the raw mark, final mark and feedback, sets the status to Marked, records an audit entry with the reason "Initial mark" (UC22), and adds the submission to the Head of Department's moderation queue. 8. The system confirms and shows the moderation status "Awaiting moderation".
Alternative flows	A1 On time (step 5). Days late is 0 and the final mark equals the raw mark. A2 One to three days late (step 5). The penalty is applied and the final mark is the raw mark less the penalty, never below 0. The Lecturer cannot edit the penalty. A3 More than three days late (step 5). The final mark is 0, the raw mark is kept and shown, and feedback is still required. A4 Re-marking a returned submission (step 1). The system shows the Head of Department's reason for returning it. At step 3 the Lecturer also enters a reason for the new mark, which is used in the audit entry. The Returned flag is cleared at step 7. A5 Cancel (at step 6). Nothing is saved.
Exception flows	E1 Mark out of range (step 4). The system shows an error and returns to step 3. E2 Empty feedback (step 4). The system shows an error and returns to step 3. E3 Submission cannot be marked (step 1). Its status is Moderated or Released. The system refuses and changes nothing. E4 Not permitted (step 1). The Lecturer is not assigned to the course. The server rejects the request. E5 Stale data (step 7). The submission changed after it was displayed. The system rejects the save and reloads the submission. E6 Database error (step 7). The transaction is rolled back and the system reports the error.
Postconditions	Success: the status is Marked. The raw mark, days late, penalty, final mark and feedback are stored. An audit entry exists. The submission is in the moderation queue and is not visible to the student. Failure: stored data is unchanged.
Business rules	BR3, BR5, BR6, BR7, BR8, BR9
Requirements	FR13, FR18, FR19, FR23, NFR08, NFR09

Use case UC17: Moderate and release marks

Primary actor	Head of Department
Stakeholders and interests	Principal: marks are checked before any student sees them. Head of Department: sees every mark in the department before release. Lecturer: approved marks are released promptly. Student: receives accurate marks and feedback.
Trigger	The Head of Department opens the moderation queue for a course in their department.
Preconditions	P1 The Head of Department is logged in and is assigned to a department. P2 At least one submission in the department's courses has status Marked.
Main flow	1. The Head of Department selects a course and an assignment in their department. 2. The system lists the Marked submissions (UC16), showing the Lecturer, raw mark, days late, penalty, final mark and feedback. 3. The Head of Department reviews them, selects submissions and chooses Approve. 4. The system sets the status of each selected submission to Moderated and records who approved it and when. 5. The Head of Department chooses Release for the Moderated submissions, either the selected ones or all Moderated submissions of the assignment. 6. The system checks that each selected submission is Moderated and has non-empty feedback. 7. The system sets each submission to Released and records who released it and when, one transaction for each submission.

	8. The system reports how many were released. Students can now see their marks and feedback.
Alternative flows	A1 Adjust mark (UC18, at step 3). The Head of Department chooses Adjust and enters a new raw mark and a reason. The system recalculates the final mark (UC21), saves it and records an audit entry with the old and new raw mark, who, when and the reason (UC22). The submission is then approved as in step 4. A2 Return for re-marking (UC19, at step 3). The Head of Department enters a reason. The submission stays Marked and flagged Returned, and the Lecturer can see the reason. It is not approved. A3 Release later. The Head of Department stops after step 4 and carries out steps 5 to 8 in a later session. A4 Release one mark. At step 5 the Head of Department selects a single Moderated submission.
Exception flows	E1 Outside the department (step 1). Courses of other departments are not offered, and a direct request for one is rejected. E2 Invalid adjustment (A1). The reason is empty or the raw mark is outside the range. The system shows an error and changes nothing. E3 Not eligible for release (step 6). A selected submission is not Moderated or has empty feedback. The system skips it and reports the reason, and releases the others. E4 Stale data (steps 4 and 7). A submission changed after the list was loaded, for example it was re-marked. The system refuses the action for that submission and reloads the list. E5 Database error (steps 4 and 7). The transaction for that submission is rolled back, and the system reports which submissions failed.
Postconditions	Success: approved submissions are Moderated. Released submissions are Released and visible to the student. Every adjustment has an audit entry. Each approval and release records the Head of Department and the time. Failure: submissions that failed are unchanged.
Business rules	BR3, BR5, BR6, BR7, BR8, BR9
Requirements	FR19 to FR23, FR25, FR26, NFR08, NFR09

The use case column of the requirements table and the requirements column of the catalogue are the first links of the traceability matrix, which will be extended in Phases 3 to 5 with design elements, source files and test cases and placed in the appendices.